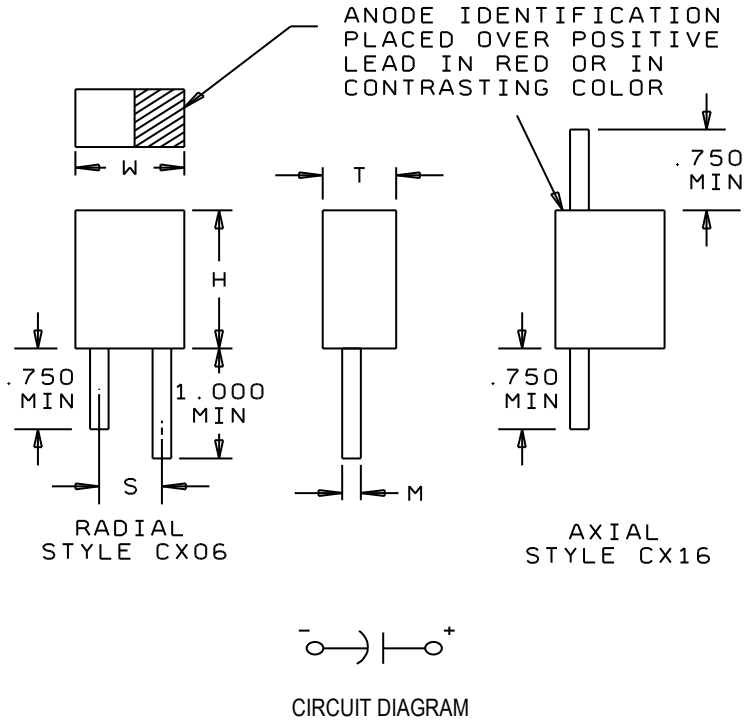
INCH-POUND
MIL-PRF-49137/6F
w/AMENDMENT 1
23 April 2021
SUPERSEDING
MIL-PRF-49137/6F
10 October 2014

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, NONHERMETICALLY SEALED, STYLES CX06 AND CX16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49137.



CIRCUIT DIAGRAM

FIGURE 1. Styles CX06 and CX16 capacitors.

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Case size	Dimensions				
	T	W	H	M	S
	Max	Max	Max	±.002	
A	.040	.050	.100	.007	.030 ± .015
B	.040	.070	.125	.010	.050 ± .015
C	.070	.120	.165	.010	.100 ± .020
D	.075	.185	.225	.010	.150 ± .020
E	.110	.220	.290	.016	.180 ± .025
F	.130	.230	.310	.016	.200 ± .025
G	.150	.375	.475	.016	.300 ± .025

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.040	1.02	.130	3.30	.230	5.84
.007	0.18	.050	1.27	.150	3.81	.290	7.37
.010	0.25	.070	1.78	.165	4.19	.300	7.62
.015	0.38	.075	1.90	.180	4.57	.310	7.87
.016	0.41	.100	2.54	.185	4.70	.375	9.52
.020	0.51	.110	2.79	.200	5.08	.475	12.06
.025	0.64	.120	3.05	.220	5.59	.750	19.05
.030	0.76	.125	3.18	.225	5.72	1.000	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. The positive lead on CX06 shall be at least .125 inch (3.18 mm) longer than the negative lead.

FIGURE 1. Styles CX06 and CX16 capacitors - Continued.

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TABLE I. Styles CX06 and CX16 characteristics.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C)	Nominal capacitance	Dissipation factor (max)	DC leakage (max) +25°C	Case size
	Volts	µF	%	µA	
CX - 6A474 -	2	.47	10	.5	A
CX - 6A225 -	2	2.2	10	.5	B
CX - 6A106 -	2	10.0	10	.5	C
CX - 6B155 -	3	1.5	10	.5	B
CX - 6B685 -	3	6.8	10	.5	C
CX - 6B226 -	3	22.0	10	1.0	D
CX - 6B227 -	3	220.0	15	9.0	G
CX - 6C334 -	4	.33	10	.5	A
CX - 6C105 -	4	1.0	8	.5	B
CX - 6C475 -	4	4.7	8	.5	C
CX - 6C156 -	4	15.0	8	1.0	D
CX - 6C476 -	4	47.0	8	2.0	E
CX - 6C686 -	4	68.0	8	3.0	F
CX - 6D224 -	6	.22	10	.5	A
CX - 6D684 -	6	.68	6	.5	B
CX - 6D335 -	6	3.3	6	.5	C
CX - 6D106 -	6	10.0	6	1.0	D
CX - 6D336 -	6	33.0	6	2.0	E
CX - 6D476 -	6	47.0	6	3.0	F
CX - 6D157 -	6	150.0	10	9.0	G
CX - 6F154 -	10	.15	10	.5	A
CX - 6F474 -	10	.47	6	.5	B
CX - 6F225 -	10	2.2	6	.5	C
CX - 6F685 -	10	6.8	6	1.0	D
CX - 6F226 -	10	22.0	6	2.0	E
CX - 6F336 -	10	33.0	6	3.0	F
CX - 6F107 -	10	100.0	8	9.0	G
CX - 6H104 -	15	.10	10	.5	A
CX - 6H334 -	15	.33	6	.5	B
CX - 6H155 -	15	1.5	6	.5	C
CX - 6H156 -	15	15.0	6	2.0	E
CX - 6H226 -	15	22.0	6	3.0	F
CX - 6H686 -	15	68.0	8	9.0	G
CX - 6J104 -	20	.10	6	.5	B
CX - 6J154 -	20	.15	6	.5	B
CX - 6J224 -	20	.22	6	.5	B
CX - 6J105 -	20	1.0	6	.5	C
CX - 6J335 -	20	3.3	6	1.0	D
CX - 6J475 -	20	4.7	6	1.0	D
CX - 6J106 -	20	10.0	6	2.0	E
CX - 6J156 -	20	15.0	6	3.0	F
CX - 6J476 -	20	47.0	8	9.0	G
CX - 6K684 -	25	.68	6	.5	C
CX - 6K225 -	25	2.2	6	1.0	D
CX - 6K685 -	25	6.8	6	2.0	E
CX - 6K106 -	25	10.0	6	3.0	F
CX - 6K336 -	25	33.0	6	9.0.	G

See footnote at end of table.

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TABLE I. Styles CX06 and CX16 characteristics - Continued.

PIN 1/	DC rated voltage (+85°C)	Nominal capacitance	Dissipation factor (max)	DC leakage (max) +25°C	Case size
	Volts	µF	%	µA	
CX - 6M104 -	35	.10	6	.5	C
CX - 6M154 -	35	.15	6	.5	C
CX - 6M224 -	35	.22	6	.5	C
CX - 6M334 -	35	.33	6	.5	C
CX - 6M474 -	35	.47	6	.5	C
CX - 6M684 -	35	.68	6	1.0	D
CX - 6M105 -	35	1.0	6	1.0	D
CX - 6M155 -	35	1.5	6	1.0	D
CX - 6M225 -	35	2.2	6	2.0	E
CX - 6M335 -	35	3.3	6	2.0	E
CX - 6M475 -	35	4.7	6	2.0	E
CX - 6M685 -	35	6.8	6	3.0	F
CX - 6M106 -	35	10.0	6	9.0	G
CX - 6M156 -	35	15.0	6	9.0	G
CX - 6M226 -	35	22.0	6	9.0	G

1/ Complete PIN will include additional symbols to indicate style and capacitance tolerance.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Polyester sleeve and epoxy end filled or plastic encapsulated.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55° to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Capacitance tolerance: ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Terminal strength: In accordance with MIL-STD-202-211.

Test condition A: One-half pound for case sizes A and B, 1 pound for case sizes C and D, and 2 pounds for case sizes E, F, and G.

Test condition C: One-half pound applied force - 3 bends.

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Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°C):

DCL: See table I.
Cap.: Within tolerance specified in table I.
DF: See table I.

Step 2 (-55°C):

ΔCap.: Within ±15 percent of step 1 measured value.
DF: 200 percent of table I.

Step 3 (+25°C):

DCL: See table I.
ΔCap.: Within ±10 percent of step 1 measured value.
DF: See table I.

Step 4 (+85°C):

DCL: 10 times +25°C limit (see table I).
ΔCap.: Within ±15 percent of step 1 measured value.
DF: 150 percent of table I.

Step 5 (+25°C):

DCL: See table I.
ΔCap.: Within ±10 percent of step 1 measured value.
DF: See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL: See table I.
ΔCap.: Within ±10 percent of initial value.
DF: See table I.

Life (1,000 hours at +25°C): In accordance with MIL-STD-202-108.

DCL: 200 percent of table I.
ΔCap.: Within ±10 percent of initial measured value.
DF: See table I.

Resistance to solvents: Not applicable.

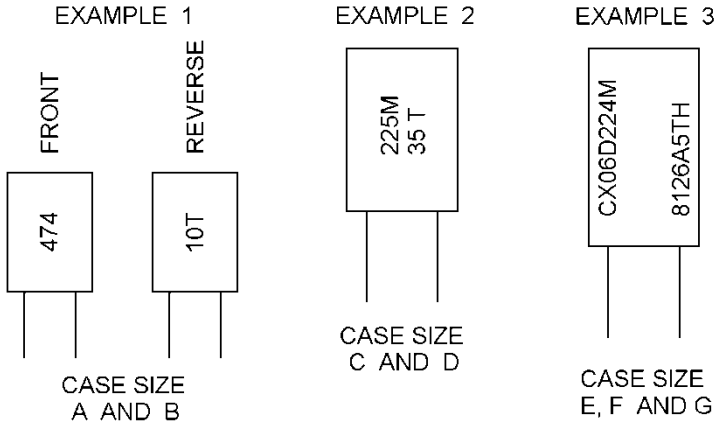
Marking: In accordance with MIL-PRF-49137. The following details and exceptions shall apply:

Case sizes A and B: Shall be marked as shown in example 1. The front shall be marked with the capacitance value coded in picofarads and the reverse shall be marked with the dc voltage and the manufacturer's trademark. At the option of the manufacturer, color coding may be used to mark the capacitance, dc rated voltage, and the manufacture's trademark.

Case sizes C and D: Shall be marked as shown in example 2. The first line shall contain the capacitance in picofarads and the capacitance tolerance. The second line shall contain the dc voltage and the manufacturer's trademark. At the option of the manufacturer, the format shown in example 3 may be used for case sizes C and D.

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Case sizes E, F, and G: Shall be marked as shown in example 3. The first line shall contain the part number and the second line shall contain the date code (year and week), lot symbol, and manufacturer's trademark. At the option of the manufacturer, three lines may be used to mark the date code, lot symbol, and manufacturer's trademark, as long as it is the same order as shown in example 3



Voltage groups (+85°C rated voltage):
I----------2 to 10 V inclusive
II----------15 to 35 V inclusive

Reference documents. In addition to MIL-PRF-49137, this specification sheet references the following:

MIL-STD-202-108
MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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